

Practicum No.4**Date: 30/1/2026****Market-Validation Questionnaire Design for a CSE/AIML/DSCS/DS Product****I. Practical Significance**

This practicum equips students with the ability to design a professional and application-specific market-validation questionnaire for a proposed CSE/AIML/DSCS/DS product or service. Market validation is a critical stage in early-stage venture creation, allowing founders to assess customer needs, willingness to adopt, perceived value, usability expectations, pricing sensitivity, and overall product-market fit. By performing this exercise, learners apply principles of user research, customer discovery, and evidence-based entrepreneurship within computing and AI-centric domains.

The practicum enhances skills in identifying appropriate questions, selecting correct scales, structuring user-centric survey flows, and aligning them with real digital product requirements such as AI automation tools, cybersecurity platforms, cloud-native services, predictive analytics engines, developer tools, or intelligent applications.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a pitch deck, project proposals for small/medium scale startups logically and scientifically.
- a) **Use** technically robust survey tools for software engineering, cloud-native systems, and intelligent computing sectors. (CDO)
- b) **Interpret** target customer clusters for SaaS applications, AI-powered automation tools, and data-driven service platforms. (CDO)
- c) **Evaluate** commercial viability of CSE products such as AI models, software applications, analytics platforms, and cloud-native solutions. (PDO)
- d) **Practice** professional ethics in data collection, system design, and user interaction within digital ecosystems. (ADO)

III. Relevant CO(s)

- **CO2. Validate** feasible business ideas using alternative entrepreneurial methods.

IV. Practical Outcome (PrO)

- **PrO4 – Design** a comprehensive questionnaire for **the identified** CSE, AIML, DSCS & DS start-up product/service to validate the market need.

V. Related ADO(s) (keep as such)

- a) **Practice** professional ethics.
- b) **Function** effectively as a team member/leader.
- c) **Practice** integrity and honesty.

VI. Minimum Underpinning Theory

Market validation is the process of verifying whether a proposed solution addresses a real customer problem and whether the market is willing to adopt and pay for it. A well-designed questionnaire ensures:

- Identification of customer personas.
- Verification of the significance of the problem.
- Understanding of existing alternatives.
- Insights into user expectations and decision factors.
- Preliminary pricing/demand estimation.

Common question categories include:

- Screening questions
- Problem understanding
- Current solution behavior
- Feature expectations
- Benefit perception
- Adoption likelihood
- Pricing expectations
- Open-ended insights

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VIII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For research and documentation	1 per team
2	Internet access	For identifying trends and problems	As required
3	Survey platform	Google Forms, Typeform, MS Forms	1 per team
4	Reference product data	Competitor surveys, market insights	1 per team

IX. Safety Precautions

- a) **Avoid** collecting personally sensitive data (address, phone number).
- b) **Maintain** confidentiality of respondent information.
- c) **Use** the collected data only for academic/educational purposes.
- d) **Obtain** consent before collecting responses.

X. Procedure

1. **Select** an application-specific CSE/AIML/DSCS/DS product.
2. **Identify** the target customer segment.
3. **Draft** question categories based on validation objectives.
4. **Prepare** 15–25 structured questions (MCQ, Likert scale, ranking, open-ended).
5. **Review** for neutrality, clarity, and relevance.
6. **Document** the final questionnaire.
7. **Present** your findings in a report or slide deck.

XI. Actual Resources Used *(To be filled by the students)*

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	HP	Used for questionnaire design	1	Personal device
2.	Internet Access	Airtel WiFi	Used for research and survey distribution	1	Stable connection
3.	Survey Platform	Google Forms	Used to create and collect responses	1	Free tool

XII. Actual Procedure Followed

1. The product idea **Digital Attendance and Productivity Tracker** was selected for market validation.
2. The **target users were identified as college students and teachers** who manage attendance and academic activities.
3. The **purpose of the questionnaire** was defined to understand problems in attendance management and the need for a digital tracking system.
4. Different **question categories were prepared**, such as user details, current attendance methods, problems faced, expected features, and usage preference.
5. A **questionnaire containing 15–20 questions** was designed using **Google Forms**.
6. The questions were reviewed to ensure they were **clear, simple, and relevant to the project idea**.
7. The survey form was shared with **students and faculty members**, and responses were collected for analysis.

XIII. Observations and Data

S. No.	Survey Question / Parameter Observed	Data/ Remarks
1	Current method used for attendance tracking	Many respondents use manual registers or simple digital tools
2	Difficulty in managing attendance records	Around 70% of respondents reported that managing attendance manually is time-consuming
3	Interest in automated attendance system	Majority of respondents showed interest in a digital attendance tracker
4	Expected features in the system	Automatic attendance recording, productivity tracking, and performance reports were highly preferred
5	Willingness to use the system	Most students and teachers said they would use the system if it is simple and reliable

XIV. Results/Observations

1. The survey results indicate that many students and teachers still rely on **manual attendance systems**, which can be inefficient and difficult to manage.
2. Many respondents expressed interest in using a **Digital Attendance and Productivity Tracker** that can automatically record attendance and generate useful productivity reports.
3. Respondents also suggested that features like **automated reports, notifications, and easy data access** would make the system more useful.

XV. Interpretation of Results

1. The responses show that there is a clear need for a more efficient system to manage attendance and track productivity in educational environments.
2. A digital system that automates attendance recording and provides productivity insights could reduce manual work and improve monitoring for both students and teachers.
3. The results also indicate that users prefer a system that is simple to use, reliable, and accessible through digital devices.

XVI. Conclusions

From this survey, it can be concluded that the Digital Attendance and Productivity Tracker is a useful and practical idea for educational institutions.

Many respondents showed interest in features such as automatic attendance recording, productivity monitoring, and performance reporting.

If the system is designed to be user-friendly, accurate, and accessible, it has strong potential to be adopted by students and teachers. Further improvements can include mobile application support, notifications, and data analytics features to enhance the overall functionality of the system.

XVII. Suggested Practicum Related Questions

1. **Evaluate** how each section of the questionnaire supports customer discovery for AI-based or data-centric solutions.
2. **Examine** the role of scaling questions in assessing perceived usefulness and adoption likelihood.
3. **Interpret** how user responses can guide feature prioritization and MVP development.
4. **Assess** the potential biases and limitations of self-reported customer data.
5. **Justify** the selection of question formats for different sections of the survey.

XVIII References / Suggestions for further reading (keep as such)

1. Venture Meets Mission: Aligning People, Purpose, and Profit to Innovate and Transform Society, Arun Gupta, Gerard George, Thomas J. Fewer, Stanford University Press, Stanford, California, 2024, ISBN: 978-1503636283
2. So You Want to Start a Business: Eight Steps to Startup Success, John B. Vinturella, Business Expert Press, New York, USA, 2024, ISBN: 978-1637425855
3. Business Planning, Franklin A. Gevurtz, Foundation Press (West Academic), St. Paul, Minnesota, 2024, ISBN: 978-1685612764

XIX Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the **process-related skills** and **product-related skills** will change depending on the PrO, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Identification of objectives	3	
2	Logical question structuring	4	
3	Ethical clarity	4	
4	Collaboration	3	
5	Technical relevance	4	
Product-related Skills: 12 Marks - 40%			
1	Quality of questionnaire design	3	
2	Alignment with startup concept	3	
3	Clarity and organization	3	
4	Submission quality	3	
Total		30	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	Neavis Rishva. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained	Name and Designation of the Faculty Member	Signature with Date
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Process-Related skills (18)	Product-Related skills (12)	Total (30)		

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